

$$A: A \xRightarrow{*} \epsilon$$

$$A \rightarrow B'C'$$

$$B \xRightarrow{*} \epsilon$$

$$\hookrightarrow A \rightarrow C$$

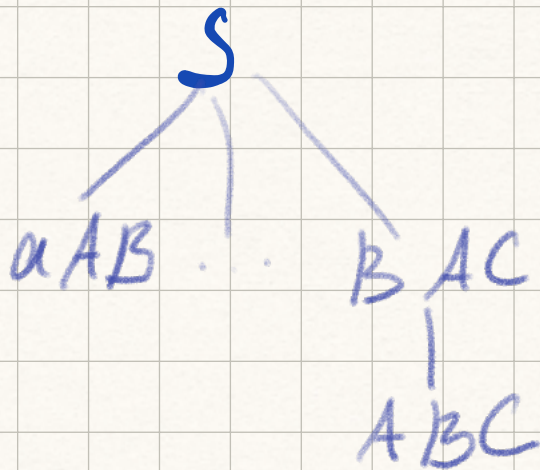
$$C \xRightarrow{*} \epsilon$$

$$A \rightarrow B$$

Лекция 4

СУК-Алгоритм LL-анализ

$$\alpha \rightarrow \beta \quad |\beta| \geq |\alpha|$$



$$A \rightarrow B \quad B \rightarrow A$$

$$w \in L(G)$$

СУК-Алгоритм

$$A \rightarrow a ; A \rightarrow BC \quad S \rightarrow \varepsilon$$

$$w = w_1 \cdot w_2 \cdot \dots \cdot w_n$$

$$w_i \in \Sigma$$

$$X_{ij} = \{ X : X \xRightarrow{*} w_i \dots w_j \}$$

$$w \in L(G) \Leftrightarrow S \in X_{1n}$$

$$0) X_{ii} = \{ X : X \xRightarrow{*} w_i \}$$

$$k) X_{i, i+k} = \{ X : \begin{array}{l} (X \rightarrow AB) \\ A \in X_{i,j} \\ B \in X_{j+1, i+k} \end{array} \}$$

$\downarrow *$
 $w_i \cdot w_j \cdot w_{i+k}$

индукция по k

Handwritten practice of the letter 'a' in cursive, showing five examples labeled 1 to 5.

$$w_2 w_3 w_4 = bba$$
$$X_{i,j} = \{x: x \stackrel{*}{\rightarrow} bba^j\}$$
$$S \in \mathcal{X}_{1,n}$$

for $j := 1$ to $n-1$

for $S \rightarrow AB$:

if $(A \in X_{1,j}) \wedge (B \in X_{j+1,n})$:

break // $S \rightarrow AB$

 $O(n)$

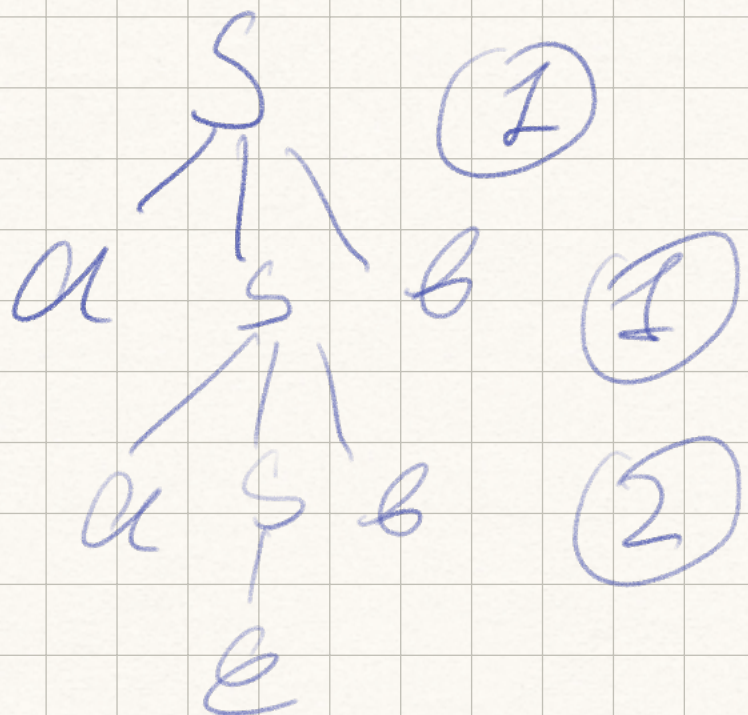
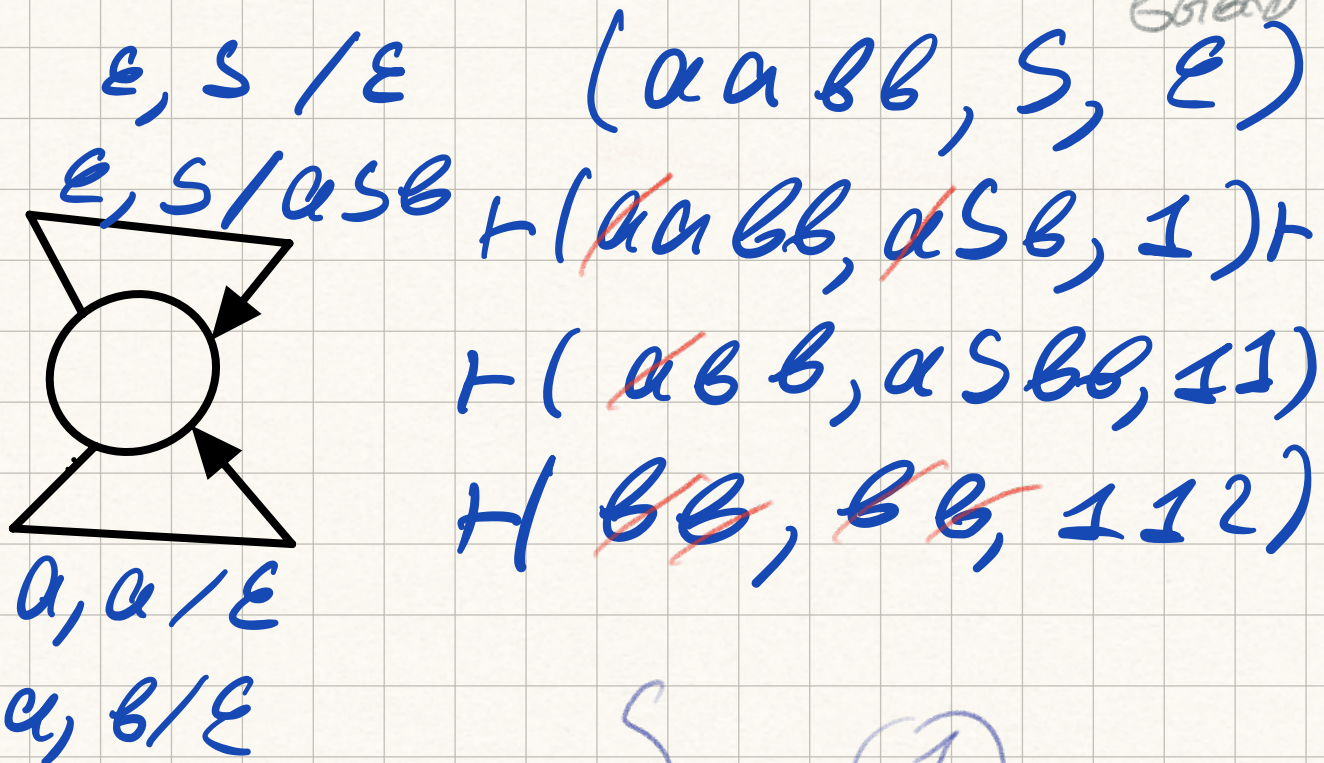
$$X_{i, i+k} \stackrel{O(n)}{=} \{ X : \begin{array}{l} (X \rightarrow AB) \\ A \in X_{i,j} \\ B \in X_{j+1, i+k} \end{array} \mid j \in [i, i+k) \}$$

$$\# \{ X_{i,j} \} = \Theta(n^2)$$

$$T(n) = O(n^3)$$

$S \rightarrow aSb^1 / \epsilon^2$
 $R(a, S) = 1$
 $R(b, S) = 2$

$S \Rightarrow_1 aSb \Rightarrow_1 aaSbb \Rightarrow_2 aa bb$



$$S \Rightarrow_L p_1 A_1 d_1 \Rightarrow_L p_2 A_2 d_2 \Rightarrow_L \dots$$

$$p_i \in \Sigma^*, A_i \in N \Rightarrow_L p_n = w$$

$$d_i \in (\Sigma \cup N)^*$$

$$p_i A_i d_i \Rightarrow_L p_i B_i d_i$$

$$A_i \rightarrow B_i$$

$$w = p_i a s_i$$

разделитель

$$B_i = R(A_i, a)$$

$$Rule(A_i, a) = \begin{cases} A_i \rightarrow B_i \end{cases}$$

если грам. не
попал

ошибка,
если слово
не ввб.

* Грамматика
полная

$$\underline{S' \rightarrow S\$} \quad \$ \in \Sigma$$

$$\left(G \rightarrow G' \right)$$

↑
над Σ

$$p_i A_i d_i \Rightarrow_L p_i B_i d_i$$

$$w = p_i a s_i \quad A_i \rightarrow B_i$$

$$\text{I. } B_i \xRightarrow{*} a u \quad (s_i = a v)$$

$$\text{II. } \begin{cases} B_i \xRightarrow{*} \varepsilon \\ d_i \xRightarrow{*} a v \quad (s_i = v x) \end{cases}$$

$$\text{First}(a) = \{ a \mid a \xRightarrow{*} a u \} \cup \{ \varepsilon \mid a \xRightarrow{*} \varepsilon \}$$

$$S' \rightarrow S \$$$

↗ Новая
 аксиома

↖ маркер конца
 слова $\$ \notin \Sigma$

$$\text{Follow}(A) = \{a \mid S' \xRightarrow{*} P_i A \beta_i \xRightarrow{*} w \$ y\}$$

$$A \in N$$

$$a \in \Sigma \cup \{ \$ \}$$

Характеристики грамматики ~
- приведённые

$$\text{I. } \forall A \in N : A \xRightarrow{*} u \in \Sigma^*$$

$$\text{II. } \forall A \in N : S' \xRightarrow{*} \alpha A \beta \xRightarrow{*} w \$$$

Критерий $LL(1)$ - грам.

$$\text{I. } a \in \text{First}(A_i) \Rightarrow$$

$$\Rightarrow \exists ! A_i \rightarrow \beta_i : a \in \text{First}(\beta_i)$$

$$\text{II. } \epsilon \in \text{First}(A_i) \Rightarrow$$

$$\Rightarrow \text{First}(A_i) \cap \text{Follow}(A_i) = \emptyset$$

$$p_i A_i d_i \Rightarrow_L p_i \beta_i d_i$$

$$w = p_i \alpha s_i \quad A_i \rightarrow \beta_i$$

$$\text{I. } \beta_i \stackrel{*}{\Rightarrow} \alpha u \quad (s_i = u v)$$

$$\text{II. } \beta_i \stackrel{*}{\Rightarrow} \varepsilon$$

$$(d_i \stackrel{*}{\Rightarrow} \alpha v \quad (s_i = v x))$$

$S \rightarrow aS^{(1)}b \mid \epsilon^{(2)} \quad S' \rightarrow S\$$

First S
 $= \{a, \epsilon\}$

Follow S
 $= \{b, \$\}$

	a	b	$\$$
S	$S \xrightarrow{(1)} aSb$	$S \xrightarrow{(2)} \epsilon$	$S \xrightarrow{(2)} \epsilon$

$(\underline{aa}bb\$, \underline{S}\$, \epsilon) \vdash$

$(\cancel{aa}bb\$, \cancel{a}Sb\$, 1) \vdash$

$\vdash (\cancel{a} \underline{bb}\$, \cancel{a} \underline{S}bb\$, 11) \vdash$

$\vdash (\cancel{bb}\$, \cancel{bb}\$, 112)$

1. Проведение зр.

2. Выход. First

3. Выход Follow

1. Приведение гр.

дост.: $\{A: S \xRightarrow{*} \alpha A \beta \xRightarrow{*} w\}$

полновитые $\{A: A \xRightarrow{*} u\}$

сепарационные

- ① удаляем сеп. нет.
- ② к-лем. не дост и удаляем.

$$V_0 = \Sigma \quad \textcircled{1}$$

$$V_i = \{ A : \exists A \rightarrow \alpha \alpha \in (V_{i-1})^* \}$$

$$V_k \quad k = |N|$$

②

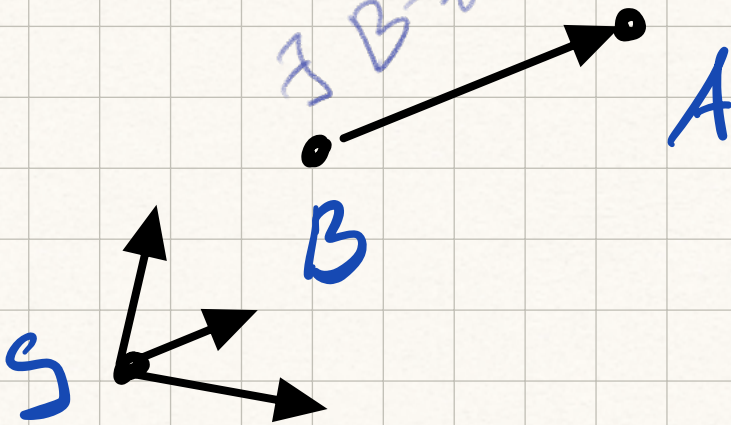
$$V_0 = \{S\}$$

$$V_i = \{A :$$

$$\exists B \in V_{i-1} \mid \\ B \rightarrow \alpha A \beta \in P\}$$

$(V_{i-1}) \cup$

$\exists B \rightarrow \alpha A \beta \in P$



$$\text{First}(X_1 \cdot X_2 \cdot \dots \cdot X_n) =$$

$$X_i \in N \cup \Sigma$$

$$= \text{First}(X_1) \mid \epsilon \in X_1 \cup$$

$$\cup (\text{First}(X_2) \mid \epsilon \in X_2) [\epsilon \in \text{First}(X_1)]$$

$$\cup \dots$$

$$\cup (\text{First}(X_n) \mid \epsilon \in \text{First}(X_{n-1})]$$

$$\{y_{cu}\} = \begin{cases} 1, & \text{yes. верно} \\ 0, & \text{yes. неверно} \end{cases}$$

$$F_i(a) = \{a\} \quad \forall i, a \in \Sigma$$

$$F_0(A) = \begin{cases} \{\epsilon\}, & A \rightarrow \epsilon \in P \\ \emptyset, & A \rightarrow \epsilon \notin P \end{cases}$$

$$F_i(A) \quad i \geq 1 :$$

$$\forall A \rightarrow x_1 x_2 \dots x_n$$

$$F_i(A) \leftarrow F_{i-1}(A) \cup$$

основная
чл-во,
добавляем
элементы

$$\cup F_{i-1}(x_1) \setminus \{\epsilon\} \cup$$

$$\{ \epsilon \in F_{i-1}(x_1) \} ($$

$$(F_{i-1}(x_2) \setminus \{\epsilon\}) \cup$$

$$\cup \{ \epsilon \in F_{i-1}(x_2) \} ($$

остаточная

$$F_i(A) = F_{i-1}(A)$$

для всех
A.

$$\left(\begin{array}{l} (S \xRightarrow{*} P A \beta) \\ \beta \xRightarrow{*} \epsilon \end{array} \right)$$

$$\Leftrightarrow \epsilon \in \text{Follow}(A)$$

$$B \rightarrow \alpha A \beta$$

Donna

$$\text{Follow}(S) = \$$$

1. $\text{Follow}(A) \supseteq \text{First}(\beta) \mid \alpha \in \epsilon$
2. $\epsilon \in \text{First}(\beta) \Rightarrow$
 $\Rightarrow \text{Follow}(A) \supseteq$
 $\supseteq \text{Follow}(B)$

$$F_0(S) = \{ \$ \}$$

$$F_0(A) = \emptyset, A \neq S$$

$$\text{While } F_i(A) \neq F_{i-1}(A)$$

